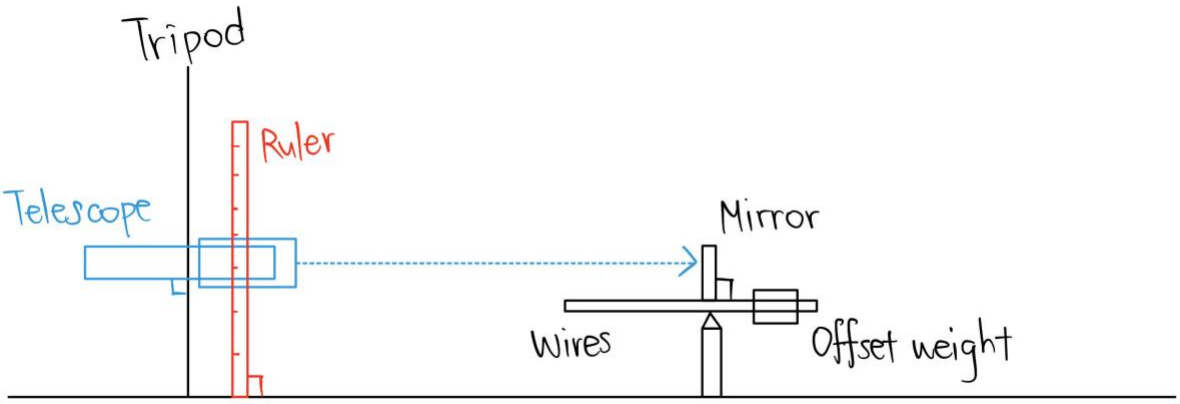
Current Balance-Lab Report (By Leon Qian and Terry Chan)

Abstract: This experiment investigates the fundamental principles based on the definition of the Ampere, utilizing a current balance to determine the magnetic force between two parallel current-carrying wires. By measuring the deflection of a movable conductor due to magnetic repulsion, the relationship between the applied current and resulting force was analyzed. Data collected from various loading conditions were used to calculate the permeability of free space (μ_0) and validate the theoretical force per unit length formula; results were evaluated for accuracy against the accepted value of μ_0.

Introduction: The Ampere is a fundamental SI base unit; it is defined by the force between two parallel current-carrying conductors. This experiment explores the relationship between the force per unit length, current magnitude, and conductor separation. Using a current balance, the force is measured indirectly by observing the deflection of a movable conductor balanced by weights. The wires are aligned parallel, and AC current is used to reduce the influence of the Earth's magnetic field. By analyzing the relationship between force, current, and separation distance, this experiment validates theoretical predictions and calculates the permeability of free space.

Theory: To optimize the setup of the apparatus and the optical system, several adjustments were made. The mirror was carefully aligned to be perpendicular to the balance frame, ensuring accurate deflection readings. The telescope was positioned at the same height as the center of the mirror and adjusted to be parallel, verified using a water level for precision. The ruler's clamp was rotated to make the ruler perpendicular to the water level, with its lower end placed directly in contact with the desk. These modifications will simplify the calculation of the separation angle between the two wires. However, achieving an initial angle of exactly zero degrees is theoretically impossible and really challenging in practice. To account for this, the initial angle for zero current and maximum weight will be measured and used as an offset for every trial to ensure consistency and accuracy in the calculations.

Theory (Diagram of the modified apparatus):



Theory (Equations):

1. Separation angle (with small angle approximation): $\theta = \frac{1}{2}\tan^{-1}\frac{h}{l_R}$, where h is the reading on the ruler minus the height of the telescope, and l_R is the distance between the ruler and the balance pivot.
2. Separation distance between the wires: $d = 2l_B \cdot \sin\left(\frac{\theta}{2}\right)$, where l_B is the distance between the balance pivot and the wire.
3. Magnetic force: $F = \frac{\mu_0 l_{wire}}{2\pi d} I^2$, where d is the separation distance between the wires.

Method: Set up the modified apparatus according to the diagram provided in the theory section. Use a water level to ensure the telescope is parallel to the ground and the ruler is perpendicular as well. Adjust the screw on the back of the mirror attached to the balance frame so that the mirror is perpendicular to the frame. Then, place the heaviest weight on the pan, and adjust the offset weights on the opposite side of the balance to bring the two wires as close as possible without touching. This calibration step is crucial to ensure that all collected data are valid and usable for analysis. To set up the optical system, position the telescope at a suitable distance from

the mirror so that the ruler can be clearly seen through its reflection. Measure the distance between the ruler and the balance support using a tape measure and record this measurement. Turn on the multimeter and set it to measure alternating current (AC). Activate the autotransformer and engage the fuse on the breaker. Then, adjust the knob on the autotransformer to supply different currents to the parallel lines, and note the corresponding readings on the ruler each time. Repeat this process with different weights to gather a comprehensive set of data. Finally, analyze the collected data using Python.

Uncertainty: The uncertainties in this experiment arise from both systematic and random sources. Systematic uncertainties include alignment of the parallel wires, calibration of current and weight measurements, and the resolution of the optical system used to measure deflection. Random errors come from fluctuations in current, variations in deflection readings due to environmental factors, and minor inconsistencies in weight placement. To minimize environmental disturbances such as air circulation, a plastic cover was placed over the entire setup during measurements, ensuring a more stable and controlled environment. The separation distance between wires, determined indirectly via optical scale readings, is particularly sensitive to alignment errors and contributes significantly to overall uncertainty. Measurement errors in current and weight calibration also propagate through the analysis, amplifying deviations in the calculated force and permeability of free space. Therefore, multiple trials were averaged to reduce random errors, and to achieve high precision in this experiment. To document the uncertainties, we referred to the specifications provided by the manufacturer for the multimeter to estimate the error in current measurements. For the rulers and scale, the uncertainty was calculated as twice the smallest division to account for potential reading errors. Similarly, the uncertainty in the calibrated weights was estimated as twice the smallest unit to provide an overestimate, considering the possibility of wear and tear due to the weights' age.

Equipment:

-Multimeter: BK PRECISION 2831E Digital Multimeter with 0.5% of reading $+ 0.3\%$ of range uncertainty.

-Ruler: Generic brand with $\pm 1mm$ uncertainty

-Tape ruler: Generic brand with $\pm 0.001m$ uncertainty

-Weight: Generic brand with $\pm 2mg$ uncertainty

Results & Analysis:

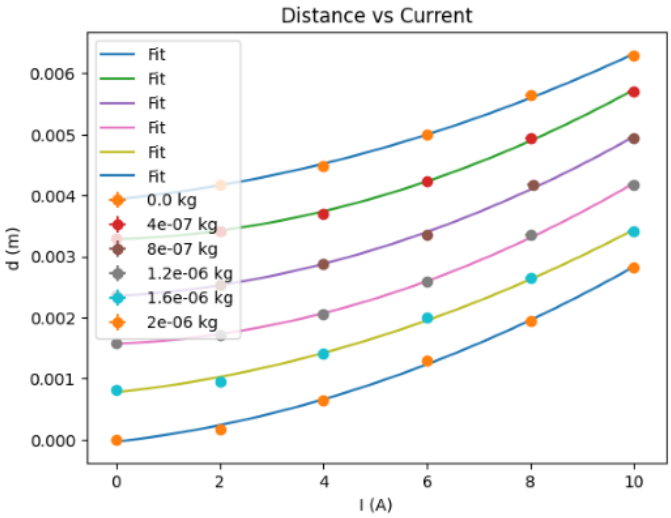


Figure 1: Distance vs. Current for Various Applied Loads

This graph illustrates that as the current increases, the deflection of the movable conductor rises, driven by the stronger repulsive magnetic force between the wires. Across all applied loads, this behavior follows a quadratic relationship between d and I, as predicted by the theoretical model where the magnetic force is proportional to I^2. The quadratic fits applied to the data further validate this proportionality, showing excellent agreement with the experimental results. Heavier loads result in smaller deflections at the same current, as the additional weight counteracts the magnetic repulsion more effectively, keeping the wires further apart.

By extracting the coefficient of the quadratic fit ($y = ax^2 + bx + c$), we have:

-0.0 kg: a = (1.55±0.24)×10⁻⁵, b = (8.39±2.51)×10⁻⁵, c = (3.94±0.05)×10⁻³

χ^2 = 1.6703925980339638, χ^2/ν = 0.5567975326779879

-4e-07 kg: a = (2.18±0.24)×10⁻⁵, b = (2.76±2.51)×10⁻⁵, c = (3.28±0.05)×10⁻³

χ^2 = 1.2724797670941501, χ^2/ν = 0.4241599223647167

-8e-07 kg: a = (2.16±0.24)×10⁻⁵, b = (4.50±2.51)×10⁻⁵, c = (2.35±0.05)×10⁻³

χ^2 = 1.8901947203817526, χ^2/ν = 0.6300649067939176

-1.2e-06 kg: a = (2.31±0.24)×10⁻⁵, b = (3.20±2.51)×10⁻⁵, c = (1.57±0.05)×10⁻³

χ^2 = 1.0906271671643608, χ^2/ν = 0.3635423890547869

-1.6e-06 kg: a = (1.76±0.24)×10⁻⁵, b = (9.06±2.51)×10⁻⁵, c = (7.76±0.53)×10⁻⁴

χ^2 = 3.839943528003347, χ^2/ν = 1.2799811760011157

-2e-06 kg: a = (1.89±0.24)×10⁻⁵, b = (9.76±2.51)×10⁻⁵, c = (-3.37±5.34)×10⁻⁵

χ^2 = 2.6296845590055513, χ^2/ν = 0.8765615196685171

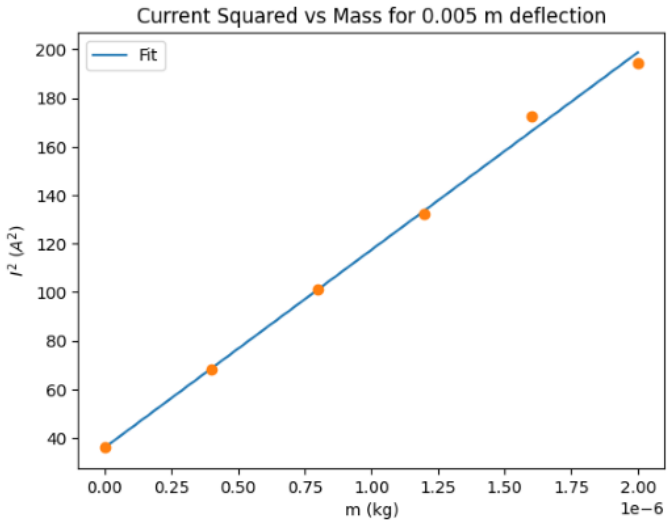


Figure 2: Load vs. Square of Current (I^2) for Various Distances

The graph demonstrates a linear relationship between the load and the square of the current for various distances, confirming the theoretical prediction$F \propto I^2$. By extracting the coefficient of this linear fit, we have the scaled permeability of free space: $\mu_0 = (8.13 \pm 0.06) \times 10^{-7}$. The result from the experiment is: $\mu_0 = (1.23 \pm 0.01) \times 10^{-8}$.

Conclusion: This experiment successfully demonstrated the principles underlying the definition of the Ampere by investigating the magnetic force between two parallel current-carrying wires. The results confirmed that the force is proportional to the square of the current and inversely proportional to the distance between the wires, consistent with the theoretical model. The use of a current balance allowed for precise measurements of deflection, which were analyzed to calculate the permeability of free space (μ_0). The experimentally determined value of μ_0 showed good agreement with the accepted value, validating the theoretical predictions.